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All Days Are Not Created Equal: Understanding Momentum by Learning to Weight Past Returns ☆

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ABSTRACT

By flexibly weighting the information contained in past realized returns, we construct a momentum strategy that outperforms and subsumes the performance of traditional stock momentum. The strategy performs well in crises and continues to work in the most recent decades. We show that the way past returns are weighted is in line with the strategy exploiting an underreaction to the information contained in realized returns, but also investigate alternative behavioral and risk-based explanations. We find that the response to earnings announcements, market-wide jumps and large individual returns realized in the formation period are most informative about future stock returns.

1. Introduction

Momentum is one of the most prominent return anomalies and has garnered considerable interest by academics and practitioners. The two seminal studies by Jegadeesh and Titman (1993) and Carhart (1997) have not only sparked a large and growing body of academic studies, but have also prompted practitioners to translate the academic evidence on return continuation into numerous trading vehicles: many hedge funds, commodity trading advisors, and mutual funds employ momentum-based strategies. Today, momentum ETFs are a low-cost alternative for investors wishing to obtain exposure to trend-following investment styles.

Despite its practical and academic success, there is still no consensus about the origins of momentum profits. Prominent explanations include a misreaction to news but both an underreaction and a delayed overreaction have been proposed. For example, in the model of Daniel et al. (1998), investors are overconfident in their own signals, leading to a delayed overreaction to news, whereas investors in the model by Barberis et al. (1998) are conservative, such that they underreact to new information. Hong and Stein (1999) propose a combination of the two theories in a single framework. As an alternative explanation, Kelly et al. (2021) show empirically that momentum profits are well-explained by time-variation in the compensation for risk in the stock market.

In this paper, we first propose a simple change to how momentum signals are generated from past realized returns, which drastically improves momentum's profitability. We then investigate possible explanations for the success of our momentum strategy. Consistent with the adage that *all days are not created equal*, we allow the weights placed on returns realized in the formation period to differ from an equal importance assumed by standard momentum. We do so by setting up a simple machine learning framework that extracts these weights from the formation period returns themselves and firm characteristics that serve as additional instruments. Importantly, we maintain the simple and well motivated structure of a *weighted* average of past realized returns as our momentum signal and coin the resulting momentum strategy *characteristic-managed momentum (CMM)*.

We start by showing that changing the way past returns are aggregated to expected return signals significantly improves the performance of momentum strategies. Not only is CMM vastly more profitable, but it is also significantly more robust. Unlike traditional momentum, CMM has worked well in recent decades, as it does not suffer from momentum crashes discussed by Daniel and Moskowitz (2016). We can show that CMM fully subsumes the performance of standard momentum. While the resulting value-weighted return of the CMM-based high-minus-low portfolio is significant within each standard momentum portfolio, we find insignificant return spreads of sorting by the traditional momentum signal, once we control for the optimized CMM signal. Keep in

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mind that the sole difference between CMM and the standard momentum strategy is in how the same daily returns realized in the formation period are weighted.

To understand why CMM is better able to extrapolate from past returns, we first show that the weights placed on past returns are sparse: on average, just two of the 231 returns in the formation period receive a weight of 30% – a 34-fold increase compared to standard momentum's equal weighting. Just 30 returns already receive more than half of the total weight, suggesting that some returns are significantly more informative than postulated by equal weights, and others significantly less informative. To understand which days are most informative about return extrapolation, we investigate if the weights are systematically elevated on certain news and event days. We find that days with particularly large returns of the stock in question are most important, followed by days with market-wide jumps. We also find that the return response to earnings and FOMC announcements are overweighted on average. Importantly, it is not simply the presence of news or market-moving events that improve momentum in the stock market. Solely basing momentum signals on the return reactions to these news and event days is insufficient to generate profitable signals that exceed the performance of standard momentum. Instead, CMM's conditional weighting function adequately extracts information about how return trends continue in the future, by combining information from the returns themselves and firm characteristics. CMM learns these patterns directly from the data, without having to preimpose upon the model which days are likely to be important for uncovering trends in stock prices.

We next seek to understand the driving forces behind our optimized momentum strategy. As is customary in the literature, we differentiate between behavioral and risk-based explanations. For the former, we devise a simple empirical test allowing us to differentiate an over- from an underreaction to information released in the formation period by investigating the correlation between absolute returns and weights. Due to the estimation of CMM, its profitability will exceed that of standard momentum if investors misreact to *large* returns realized in the formation period. We find that CMM predominantly extracts momentum signals by overweighting these large absolute returns in the formation period, which is consistent with an underreaction of investors to the information that these returns reflect. Only among this subset of stocks, which captures between 60–90% of the cross-section over time, do we find a statistically significant and economically meaningful value-weighted high-minus-low return spread.

We instead find evidence against a risk-based explanation, using the methodology of Kelly et al. (2021). We show that, unlike standard momentum, CMM is not explained by time variation in the sensitivities to systematic factors in the stock market. Instead, the conditional compensation for risk and CMM's expected return signal are largely orthogonal and both provide information about the variation in future stock returns. Furthermore, CMM is not subject to momentum crashes. In contrast, we find that the strategy specifically excels in identifying those momentum losers that outperform in the future, which avoids much of the option-like behavior of the standard momentum strategy discussed by Daniel and Moskowitz (2016).

We corroborate this evidence using the setup proposed in Goyal et al. (2024). The authors show that among various proxies for potential explanations of momentum profits, the discreteness with which information is released during the formation period following (Da et al., 2014) enjoys the strongest empirical support in international data. We find that the same measure amplifies the profitability of CMM, as additional evidence that an underreaction to news is the primary driver of the strategy's success.

We ascertain the robustness of CMM's outperformance by showing that it survives the consideration of transaction costs, yielding a Sharpe of 0.77 after accounting for the stocks' bid–ask spreads. We find that CMM performs well in international stock markets and that the same signal can be used to uncover profitable *time-series momentum* in U.S. stocks. Common stock market factors cannot explain CMM's success, and CMM outperforms alternative augmentations of momentum strategies that have been proposed in the literature.

Literature review. This paper is part of a burgeoning literature on momentum strategies, following the seminal studies by Jegadeesh and Titman (1993) and Carhart (1997). Momentum has been shown to work in many asset classes (Menkhoff et al., 2012; Asness et al., 2013; Heston et al., 2023; Li et al., 2023), exists in risk factors (Ehsani and Linnainmaa, 2022) and in returns of varying frequencies (Lou et al., 2019; Barardehi et al., 2022). Li and Liu (2022) show in the context of portfolio optimization that assets with momentum may lead to vastly different portfolio allocations, which hinge on the asset's own momentum signal. However, in recent years momentum stopped working. Green et al. (2017) show that momentum – along with many other characteristic-sorted investment strategies – ceased to generate significant returns after 2003. Furthermore, Daniel and Moskowitz (2016) show that momentum is prone to severe crashes. We propose to change how past returns are weighted to generate momentum investment signals. This seemingly simple change delivers a momentum strategy that shows great performance also after 2003 and is not subject to crashes, showcasing that past realized returns on some days are more informative than on others.

There is still an active debate about the origin of momentum profits. Prominent explanations include an under- or an overreaction to news (Daniel et al., 1998; Barberis et al., 1998). For example, in the model of Hong and Stein (1999), two types of agents consume different streams of information. One investor type impounds private news, which diffuse only gradually among them, into market prices, leading to an initial underreaction. The other investor type follows the recent trend, which leads to a delayed overreaction. In contrast, Kelly et al. (2021) show that momentum signals are a proxy for time variation in factor sensitivities. Finally, Goyal et al. (2024) compare various empirical proxies for the different explanations of momentum profits in international stock markets. By investigating the weights our momentum strategy places on returns realized in the formation period, we find compelling evidence that an underreaction to news drives its success. In adjacent empirical tests, we find corroborating evidence for this result and rule out alternative explanations.

The algorithm proposed is part of a recent shift towards using machine learning methods to predict future returns in various asset classes (Gu et al., 2020; Chen et al., 2023; Bali et al., 2020, 2023). Few other studies have investigated the usefulness of machine learning for improving return forecasts based on price trends (Jiang et al., 2023; Murray et al., 2024). Different from these studies, we retain a direct comparison to standard return-based momentum in the stock market. The sole change we propose is in the weighting function of returns realized in the formation period. Instead of the equal weights assigned by standard momentum, we allow this weighting function to change over time and across firms. With this, we also investigate the economic forces that drive our strategy's outperformance relative to standard momentum, as a central asset pricing anomaly.

Finally, our paper is related to the literature identifying important days for explaining or predicting stock returns. To this end, Savor and Wilson (2013) and Ernst et al. (2019) show that market returns are significantly higher on days with macroeconomic news releases. Liang (2003) argues that earnings announcements provide particularly important information to investors and Savor and Wilson (2016) show that stock returns are amplified around earnings announcements. Baker et al. (2021) identify days with large stock-wide absolute returns and relate them to next-day newspaper articles to understand their drivers. Our machine learning learns to weight the market's reaction to news released throughout the formation period. We find that earnings announcements, and days with large market-wide and stock-specific price moves are particularly important for the generation of momentum signals.

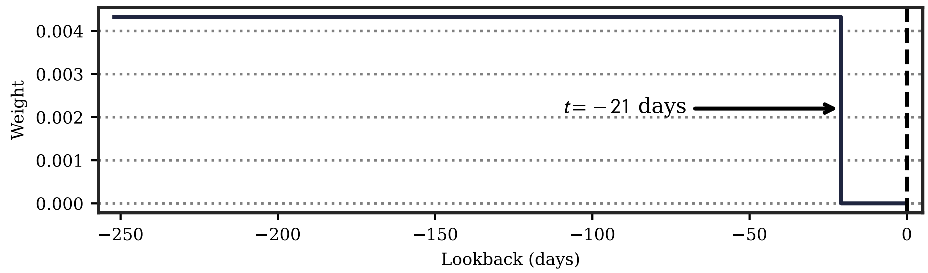

Fig. 1. Weights to Standard Momentum (SM).
The figure shows the weights assigned to past returns for the creation of the standard momentum (SM) signal. Over the past year, each of the 252 trading days is assigned an equal weight, excluding the 21 trading days of the most recent months to account for short-term return reversal effects.

2. Improving momentum signals

Traditional momentum in the spirit of Jegadeesh and Titman (1993) proposes that "winner" stocks, which have outperformed in the past, will continue to outperform "loser" stocks in the future. For this, the authors propose to compare the stocks' performance over the past year, excluding the most recent month to account for short-term return reversals. In essence, this approach assigns each of the 231 trading days of months $t-12m$ to $t-1m$ an equal importance to inform about the stock's return in month $t+1m$. The inherent return extrapolation proposed by standard momentum (SM),

$$\mathbb{E}_t^{\mathrm{SM}}[r_{i,t+1m}] = \frac{1}{231}\sum_{d=22}^{252} r_{i,t-d}, \tag{1}$$

is graphically depicted in Fig. 1. Here, $r_{i,t}$ denotes the daily log return of stock i at time t and $r_{i,t+1m}$ the one-month ahead return of i. Because we are dealing with log returns, the sorting is the same regardless of whether we use the cumulative returns over the formation period, or the average return within the formation period.

While its simplicity is appealing, the implicit assumption that the return realized on each of the 231 trading days conveys equal information about future stock returns is highly restrictive and not supported by the literature. For example, Savor and Wilson (2013) and Ernst et al. (2019) show that market returns in general are significantly amplified on days with macroeconomic releases, Knox et al. (2024) find that these same days carry significantly elevated equity premia, while Savor and Wilson (2016) and Engelberg et al. (2018) show that stock returns in general, and the returns of stock anomalies specifically, are higher around earnings announcements. Within the literature specifically targeting improvements to the momentum anomaly, we also find evidence that some days may be of higher importance: Novy-Marx (2012) shows that intermediate momentum – measured over months $t-12m$ to $t-7m$ – is more informative than using the return realized over the traditional formation period. In turn, Da et al. (2014) show that discrete information releases during the formation period amplify momentum profits, and Chen et al. (2018) show that accelerating return pattern are linked to higher momentum returns.

Following the evidence from the literature that the return reaction on some days may be more important than that on other days, we propose a small change in how past returns are aggregated into momentum-based investment signals. Consistent with the adage that *all days are not created equal*, we estimate flexible weights w using tools from machine learning, instead of assigning a fixed weight to realized returns over the past year. Importantly, we maintain the economically-interpretable structure of a linear *weighted* average of past returns:

$$\mathbb{E}_t^{\mathrm{CMM}}[r_{i,t+1m}] = \sum_{d=22}^{252} w_{i,t-d}\cdot r_{i,t-d} \tag{2}$$

subject to boundary conditions,

$$\sum_{d=22}^{252} w_{i,t-d} = 1 \qquad \text{and} \qquad w_{i,t-d} \geq 0 \quad \forall\ d \in [22, 252]. \tag{3}$$

To see how and when the strategy's expected return signal can differ from that of standard momentum, it is instructive to rewrite Eq. (2) as:

$$\mathbb{E}_t^{\mathrm{CMM}}[r_{i,t+1m}] = \underbrace{\sum_{d=22}^{252}(w_{i,t-d}-\overline{w})\cdot r_{i,t-d}}_{\text{Deviation from SM}} + \underbrace{\sum_{d=22}^{252}\overline{w}\cdot r_{i,t-d}}_{\text{Momentum signal}} \quad \text{where} \quad \overline{w} = \frac{1}{231} \tag{4}$$

The signal is composed of two parts: the standard momentum signal, and a deviation thereof. If day d^+ is expected to be more important, it will receive a weight w_{t-d^+} greater than the average weight $\overline{w}$. Because of the boundary conditions, reflecting our desire to maintain the interpretable structure of a weighted average of past realized returns, this in turn requires underweighting another day d^-, relative to the equal weights assigned by standard momentum. In other words, the setup in Eq. (4) is forced to triage between which days are expected to be particularly important for the formation of momentum signals. If an equal importance of the days in the formation period is expected to generate the most information momentum signal, the signal in Eq. (4) will simply collapse to that of standard momentum. We make use of this circumstance in Section 4 to differentiate alternative explanations for why momentum in the stock market works.

Methodology. We use both time t observable firm characteristics, $\boldsymbol{z}_{i,t}$ and the daily returns realized over the formation period, $\boldsymbol{r}_{i,t-252:t-22}$, as firm i specific instruments to allow for variation in the weights placed on past returns.[1] This allows the model to assign different weights to past realized returns for two firms with the exact same firm characteristics, depending on their profile and hence informational content of past realized returns:

$$\boldsymbol{w}_{i,t-252:t-22} = \phi(\boldsymbol{r}_{i,t-252:t-22}, \boldsymbol{z}_{i,t}). \tag{5}$$

Here, $\boldsymbol{w}_{i,t-252:t-22}$ denotes the vector of weights for the returns within the formation period. ϕ is therefore a function of observable firm characteristics $\boldsymbol{z}$ and realized daily returns $\boldsymbol{r}$ over the same formation period used for standard momentum. The sole difference to standard momentum is a departure from a fixed to a flexible weighting of past returns. The circumstance that we use firm characteristics to capture cross-sectional heterogeneity in the optimal degree of return extrapolation gives rise to the strategy's name, *Characteristic-Managed Momentum (CMM)*.

We estimate the weighting function ϕ, subject to the boundary conditions in Eq. (3), using tools from machine learning. With this, we join a growing strand of the financial literature, which uses machine learning techniques to improve the prediction of financial targets (Gu et al., 2020; Chen et al., 2023; van Binsbergen et al., 2023). For each firm available at the end of month t, we collect vector $\boldsymbol{z}_{i,t}$ of size $C = 153$, comprised of standard accounting- and market-based

[1] Using firm characteristics as instruments is frequently done in the empirical asset pricing literature, see Kelly et al. (2019), Gu et al. (2020), Kelly et al. (2021), Bali et al. (2023).

characteristics of the firm (Jensen et al., 2023). We then extract a single nonlinear feature $\hat{z}$ (of size 1) of the firm, which combines interactions between the firm's characteristics and nonlinear effects, by feeding $\boldsymbol{z}$ through a feed-forward neural network (FFN).[2] We follow (Gu et al., 2020) for the architecture and use a neural network with three hidden layers[3]:

$$\hat{z}_{i,t} = \text{FFN}(\boldsymbol{z}_{i,t}) \tag{6}$$

Generally, for an input x, a FFN alternates between a linear combination of x, and a nonlinear activation. The output of layer l is therefore:

$$x_l = \text{Activation}(W_l x_{l-1} + b_l), \tag{7}$$

with weight matrix W_l and bias b_l. The processed inputs are fed into the next layer until all layers in the FFN are exhausted. This allows the FFN to extract nonlinearities of the input characteristics, and interactions thereof.

As the final step of the FFN, we calculate a linear combination of the outputs of the last hidden layer of the FFN without adding a nonlinear activation function. The extracted feature, $\hat{z}$, is therefore a scalar. We then combine $\hat{z}$, as the description of the firm's economic conditions, with the firm's realized returns over the past twelve months, $\boldsymbol{r}_{i,t-252:t-22}$. To isolate the impact of changing only the way momentum signals are generated from past returns, we exclude the most recent month in this formation period, as in standard momentum. The combination of returns and characteristics is done by calculating their dot-product to come up with an importance Score of each of the 231 past daily returns which we denote by the vector $\text{Score}_{i,t-252:t-22}$:

$$\text{Score}_{i,t-252:t-22} = \hat{z}_{i,t} \cdot \boldsymbol{r}_{i,t-252:t-22}. \tag{8}$$

Finally, we normalize the 231 Scores per firm×month observation to obtain the weight for the past daily return at $t-d$ through a softmax function:

$$w_{i,t-d} = \frac{e^{\text{Score}_{i,t-d}}}{\sum_{d=22}^{252} e^{\text{Score}_{i,t-d}}}, \tag{9}$$

which assures that the boundary conditions outlined in Eq. (3) are fulfilled. We are thus computing an exponential average of returns realized in the formation period. Since the extracted feature $\hat{z}_{i,t}$ is a scalar for each firm×month observation, it basically acts as a scaling parameter, which governs the sparsity of the extracted weights. As we normalize the weights through a softmax function, a large absolute $\hat{z}$ amplifies differences between returns realized in the formation period, which in turn amplifies differences in the weights placed on these returns. A smaller absolute $\hat{z}$ in turn shrinks return differences, and a $\hat{z}$ of 0 collapses to equal weights in w. A positive (negative) $\hat{z}$ indicates that the model overweights large positive (negative) returns. We investigate the influence of changing levels of $\hat{z}$ for the profitability of momentum signals in Section 5.

The model is estimated by minimizing the squared distance between realized returns $r_{i,t+1m}$ and expected returns shown in Eq. (2).[4] We use the Adam optimizer (Kingma and Ba, 2014), a variant of stochastic gradient descent that converges faster and leads to better optima.

To obtain an investable CMM-based portfolio, we follow the standard momentum literature (Jegadeesh and Titman, 1993; Daniel and Moskowitz, 2016) and sort firms available at the end of month t into deciles based on $\mathbb{E}_t^{\text{CMM}}[r_{i,t+1m}]$ and using NYSE breakpoints.[5] Within each portfolio, we value-weight the included stocks. The CMM-based momentum portfolio is then the high-minus-low dollar-neutral portfolio, which we can directly compare to the standard momentum portfolio, with the sole difference being in the way past returns are aggregated to form expected return signals.

Table 1
Performance of Characteristic-Managed Momentum.
The table shows performance metrics for both the Characteristic-Managed Momentum (CMM) and standard momentum (SM) strategy. We show both, the full sample metrics spanning 1983 through 2022, as well as results for the pre- and post-2003 sample period. MDD denotes the maximum drawdown of the portfolio's value and CE($\gamma = 5$) denotes the certainty equivalent of a CRRA investor with a risk-aversion level of 5.

	CMM			SM		
	All	Pre 2003	Post 2003	All	Pre 2003	Post 2003
$\bar{r}$	0.185	0.253	0.119	0.132	0.224	0.044
	(7.65)	(8.04)	(3.92)	(2.81)	(5.55)	(0.58)
$\sigma(r)$	0.126	0.121	0.128	0.269	0.244	0.289
SR	1.465	2.090	0.934	0.490	0.918	0.153
MDD	−0.276	−0.188	−0.276	−0.803	−0.416	−0.803
Skewness	−0.180	−0.206	−0.127	−1.474	−1.127	−1.612
Kurtosis	2.420	2.902	2.334	7.010	7.034	6.583
CE($\gamma = 5$)	0.144	0.216	0.078	−0.155	0.014	−0.305

3. Investment performance

Here, we describe the data sources used to conduct our empirical analyses, and investigate the changes in the profitability of momentum-based strategies that arise from our proposed change in how momentum signals are generated.

Data. We obtain daily return data from CRSP and firm characteristics from Jensen et al. (2023) between 1973 and 2022. We consider the 153 firm characteristics studied by Jensen et al. (2023), which are collected in $\boldsymbol{z}_{i,t}$. We follow Gu et al. (2020) and iteratively expand the training sample: for the first iteration, we use data from 1973 through 1982 to estimate the model and validate its fit. The validation period comprises the last 30% of months during that period. We then use the fitted model in an out-of-sample test for all months of 1983 and 1984. Next, we expand the training and validation data set to encompass data from 1973–1984 to test the model's out-of-sample properties in 1985 through 1986. We iterate forward until all years are exhausted. This allows us to study the out-of-sample properties of CMM for the four decades between 1983 and 2022.

Profitability. Table 1 provides a comparison of characteristic-managed momentum (CMM) and standard momentum (SM). The sole difference between the two strategies is how the momentum signal – the aggregation of past daily returns – is constructed. The portfolio construction is the same, in that we show the high-minus-low return spread of value-weighted decile portfolios. CMM generates an average annualized return of 18.5%, compared to 13.2% for SM over the testing sample. The standard deviation of CMM is significantly lower at 12.6% per year vs. 26.9% for SM, which culminates in an annualized Sharpe ratio of CMM more than twice as large as that of SM (1.47 vs. 0.49). The returns of CMM are considerably more stable over time, with a maximum drawdown of −27.6%, compared to −80.3% for momentum during the financial crisis of 2008/2009. Flexibly weighting past returns circumvents the issue of momentum crashes (Daniel and Moskowitz, 2016). A less negative skewness of CMM corroborates this finding. For a moderate risk aversion of $\gamma = 5$, the certainty equivalent (CE) return

[2] We choose the Gaussian error linear unit as the nonlinear activation function, as it has better properties for stochastic gradient descent. Results using the more commonly used ReLU (Gu et al., 2020) are similar but slightly weaker.

[3] Gu et al. (2020) find that neural networks are the best-performing models, with a three-hidden-layer architecture outperforming even deeper alternatives.

[4] To facilitate the estimation of the machine learning model, we cross-sectionally normalize the target returns in each month. Importantly, this maintains the idea of differentiating winners from losers in a cross-sectional momentum strategy.

[5] Conclusions remain the same when using full sample breakpoints.

Table 2
Independent double sort.
The table shows time-series average returns of portfolios independently sorted into quintiles by the standard momentum signal (SM) and by CMM's weighted past return signal. Within each portfolio we value-weight stocks with their market capitalization. We also show high-minus-low returns. *** (**, *) denotes that the return is statistically significant at the 1% (5%, 10%) level using HAC robust standard errors.

	CMM					
SM	LOW	2	3	4	HIGH	HL
LOW	−0.94*	0.29	0.27	0.73	0.76	1.70***
2	−0.40	0.43	0.78**	0.55	0.29	0.69***
3	−0.07	0.12	0.83***	0.56	0.86***	0.93***
4	−0.20	0.36	0.57*	0.55*	0.77***	0.97***
HIGH	−0.32	0.37	0.51*	0.69**	0.49**	0.80***
HL	0.62*	0.07	0.24	−0.03	−0.27	−0.90**

of a CRRA-utility investor amounts to 14.4% per year. In contrast, the same investor would have foregone an investment in standard momentum with a negative CE, as a result of the strategy's significant volatility and drawdown.

Green et al. (2017) show that many characteristic-sorted investment strategies show subdued performance after 2003. We therefore perform a sample split to investigate if CMM continues to work in recent decades. Consistent with a performance attenuation, we find that SM generates much smaller returns since 2003 of just 4.4% per year, at an annualized Sharpe ratio of only 0.15, underperforming a buy and hold investment in the market for the same period. In contrast, CMM continues to deliver: average annualized returns amount to 11.9% with a Sharpe ratio of 0.93. While CMM's performance also attenuates after 2003, it continues to deliver profitable investment advice, highlighting that momentum-type trading and trend-following in general is still profitable today, but requires a flexible weighting of past returns to guide the decision to invest. In Table 8 we show that CMM survives the consideration of transaction costs, which we proxy by the bid–ask spreads of the stocks.

CMM subsumes standard momentum. Given the strong performance improvement of CMM compared to standard momentum, we investigate in Table 2, whether CMM subsumes the profitability of standard momentum. If it does, this highlights that changing how past daily returns are weighted not only outperforms an imposed equal importance of past returns, but that the deviation in weights from standard momentum is a strictly better way of composing momentum-based signals. Each month, we independently sort stocks into quintiles based on the standard momentum signal and CMM's signal, and record the average realized returns of the 25 portfolios, as well as the corresponding high-minus-low portfolios. We continue to value-weight stocks with their market capitalization.

We find that the CMM signal fully subsumes the profitability of the standard momentum signal. Within each of the five CMM portfolios, the resulting high-minus-low portfolio sorted by the standard momentum signal fails to generate average returns that are significant at the 5%-level. In contrast, within each of the five standard momentum portfolios, we find an economically large and statistically significant CMM-based return spread, which ranges from 0.8% per month for momentum winners (SM-HIGH) to 1.7% for momentum losers (SM-LOW). Not only do these results show the unequivocal improvement of allowing the weight placed on individual past daily returns to differ, but they also highlight that CMM particularly improves upon standard momentum by identifying those previous losers that are expected to rebound and outperform the remaining stocks. This culminates in the lowered maximum drawdown evident in both Table 1 and Fig. 4, which we analyze in more detail in Section 4.2.

Which days are important for momentum? To understand how and why CMM outperforms standard momentum, it is instructive to investigate the output of the weighting function and to see how weights w differ from an equal importance assigned by standard momentum. We first show that it is sparse on average, suggesting that some days are significantly more important than others. We then also show that earnings announcements, and days with particularly large returns are most important in the formation of CMM's signal.

Fig. 2 plots the cumulative average weight of CMM against the equal weights proposed by standard momentum. The figure can be understood as an empirical cumulative density function of weights w put on past returns. On average, just two of the 231 days in the formation period, already receive a weight of 30%, 35 days already account for more than half of the weights on average, highlighting a large degree of sparsity in the importance assigned by CMM to individual return days. We also document a large dispersion around this mean: the shaded area shows the variation within the 25th and 75th percentile of this cumulative weighting function across all firm×month observations in our out-of-sample period. The weighting of several past daily returns is significantly more sparse than the average, while several others revert back to standard momentum's equal importance.

As a natural follow-up, we investigate the average importance of certain event and news days. In each month, we perform cross-sectional regressions of the weights placed on daily returns in the formation period on different news-day indicators. We consider earnings and FOMC announcement days, the announcements of GDP and CPI figures, and days with large jumps in the stock market, which (Baker et al., 2021) identify. We also investigate if the returns of the more recent months in the formation period (months $t-6m$ to $t-2m$) or months of the latter half ($t-12m$ to $t-7m$) are more important for the generation of CMM signals, following the insight of Novy-Marx (2012) that momentum signals based on the intermediate horizon are more profitable. Finally, we investigate if days in the formation period with particularly large returns are more important, focusing on the 5% of days with the largest absolute returns. The regressions take the following form:

$$w_{i,t-d} = \alpha + \beta \cdot \mathbb{1}_{d=x} + \epsilon_{i,t-d}, \tag{10}$$

where $\mathbb{1}$ is the indicator function and x represents one of the aforementioned news and event days. That is, for each date we perform separate cross-sectional regressions of daily weights on different news-days indicators and report the time-series average of the coefficients. Reported t-values are based on HAC-robust standard errors.

The results in Table 3 produce several interesting findings. We first find that earnings announcement days are of particular importance. On average, they are weighted 115% higher than the average day, highlighting that the return response to earnings announcement carries important information for future returns.[6] Weights placed on FOMC announcement days are 15% larger than the average, whereas weights on GDP and CPI announcement days are not distinguishable from the average weight.

[6] The coefficients in Table 3 show the relative increase in the average weight.

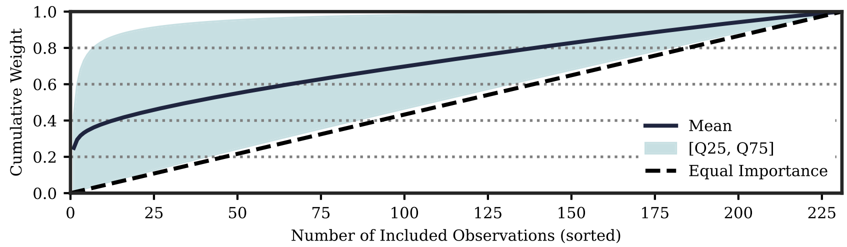

Fig. 2. Sparsity in Weights.
The figure shows cumulative weights sorted in descending order of the average weighting function ω. The shaded area highlights the variation within the 25% and 75% quantile. On average, 1% of the past observations account for 30% of the weights; 15% of observations already account for about 50% of the weights on average.

Table 3
Cross-sectional regression.
The table shows results of a cross-sectional regression analysis. For each date we perform cross-sectional regression of daily weights on different news-days indicators (cf. Eq. (10)) and report the time-series average of the coefficients. We use HAC-robust standard errors to calculate t-values. News days we consider are earnings announcement days (EA) and days with large jumps in the stock market (Baker) as defined by Baker et al. (2021). We also add to the discussion on whether the most recent-past returns (r_6_1) are more important relative to the intermediate horizon (Novy-Marx, 2012). Finally, we consider the weights on past large returns , that is, the 5% of days with the largest absolute return ($|r| > Q95$) within each time-series. To account for possible pre- and post-announcement drifts or after-hour announcements we use the average weight within a 3 day window from $t-1$ to $t+1$ days around the announcement date for EA, CPI and GDP.

	EA	Baker	FOMC	GDP	CPI	r_6_1	$\|r\| > Q95$
	Univariate regression						
Coeff.	1.15	1.27	0.15	−0.00	−0.01	0.00	5.42
t − value	(13.81)	(5.14)	(2.43)	(−0.23)	(−1.49)	(0.02)	(25.18)

In the remaining regression specifications we show that both returns in the first five months of the formation period, and the remaining six are equally important on average. Days on which the overall stock market jumps are overweighted by 127% on average, and days with a particularly large return of the stock in question are significantly more informative of future returns, being weighted 542% larger than an equal weighting. This is consistent with market microstructure theory, which postulates that price innovations are the consequence of new information (e.g. Kyle, 1985), such that larger price innovations carry more information. CMM is able to exploit the information released on these days.

In Table 11 of Appendix A we show that constructing equally-weighted momentum signals from the aforementioned news and event days is insufficient to construct a momentum strategy that systematically outperforms standard momentum and CMM, highlighting that a flexible weighting of the information contained in the return reaction to these days is vital to CMM's success.

4. Economic mechanism

Our study is chiefly concerned with understanding why momentum profits arise in the stock market. As a first step, we have therefore set up a momentum strategy that learns to optimally extract information from returns realized during the formation period. We now investigate, which explanation for momentum profits put forth in the literature best aligns with the profitability of CMM. Although momentum in the cross-section of stocks has been studied for more than thirty years, going back to the seminal paper of Jegadeesh and Titman (1993), there is still no consensus about the origins and drivers of momentum profits in that market. Possible explanations can be broadly clustered into two "camps", one relating momentum profits to investors' biases in how they react to news, the second to their risk exposure.

For example, momentum profits may be explained through the lens of an over- or underreaction to news. In the model of Daniel et al. (1998), investors are overconfident in the signals they generate themselves and will disregard conflicting public information. This pushes prices up, leading to momentum. Over time, as public information is gradually impounded into market prices, the stock's value will fall back in line with its fundamental value. Advocating for a story of underreacting to public news, Barberis et al. (1998) model investors as conservative, leading them to underweight new information. The subsequent adjustment of the stock's price to reflect its fundamental value generates a positive autocorrelation in returns. Instead, Kelly et al. (2021) show that past return-based characteristics predict future realized betas, and show that conditional risk exposure can explain momentum profits. Also favoring a risk-based explanation, Daniel and Moskowitz (2016) show that momentum profits are plagued by occasional but very severe crashes, which take decades to recover.

To assess which of these explanations best aligns with the profitability of CMM in the stock market, we first show by analyzing the weights w assigned by CMM that an underreaction to news released in the formation period is a better explanation than an overreaction to news. We then conduct a variety of additional empirical tests for risk-based and behavioral explanations, which are inspired by the literature investigating the origins of momentum profits.

4.1. Under- or overreacting to news

Which past returns are overweighted? To see how the trajectory of weights w can informs us about the relative likelihood that realized profits of CMM are the result of a systematic over- or a systematic underreaction to the news released during the formation period, assume that an important information event falls on day $t - d*$. If market participants *underreact* to this news and the news itself is positive for the firm, the realized return on $t-d*$ is too small, such that $r^{\text{realized}}_{t-d*} > 0$ and $r^{\text{realized}}_{t-d*} < r^{\text{efficient}}_{t-d*}$. If instead the news is negative for the firm, we have that $r^{\text{realized}}_{t-d*} < 0$ and $r^{\text{realized}}_{t-d*} > r^{\text{efficient}}_{t-d*}$. In both cases, the absolute realized return is closer to zero than what would have been efficient, as judged by the news component released on that day. To capture the resolution of this mispricing during the investment period, CMM should thus assign a greater weight to r_{t-d*}, as market prices are expected to align with fundamental values. As a result, CMM will capture an underreaction to news if the correlation between past absolute returns $|r|$ and the weights w assigned to them is positive. Analogously, to profit from a systematic overreaction of market participants to the news, the correlation between past absolute returns and weights will be negative. This insight provides us with a simple to use tool to understand whether trend-following profits are primarily driven by market participants under- or overreacting to news. Importantly we

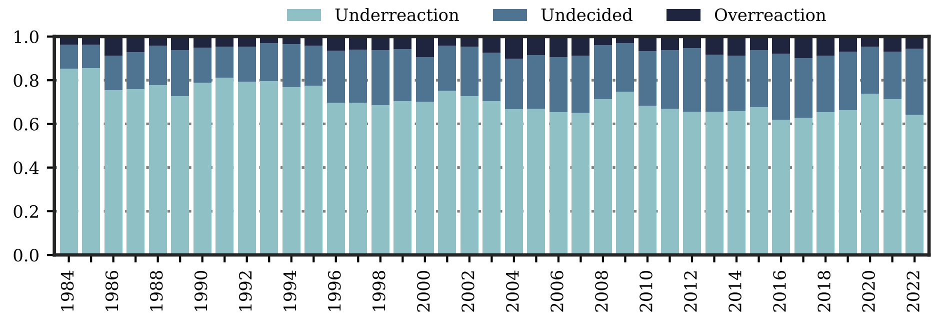

Fig. 3. Under- vs. Overreaction.
The figure shows the fraction of firm×month observations, for which the model implied correlation coefficient between the weights $w_{i,t-d}$ and absolute returns $|r_{i,t-d}|$ across all lookback days d is significantly positive or negative for different vintage years. A significantly positive correlation coefficient suggests that the model positively extrapolates past absolute returns, in line with an initial underreaction to the news component in past returns, while a negative correlation suggests the opposite. The verdict is "undecided" whenever the correlation coefficient is not statistically different from zero at the 1%-level.

can quantify the perceived share of under- vs. overreaction in the cross-section of stocks.

Of course, this line of argument requires that investors impound valuable news into market prices without delay, and that larger absolute returns reflect more information. CMM can only profit from a misreation of investors to *large* returns realized in the formation period. An alternative story by Da et al. (2014) explains momentum profits through a systematic underreaction to discrete information releases, for which Goyal et al. (2024) find strong empirical support in international data. That type of underreaction cannot be captured by the way CMM is constructed. In Table 6, we therefore investigate the relationship between CMM and alternative proxies for momentum explanations, including the information discreteness measure of Da et al. (2014).

For all firm×month observations in our sample, we compute the corresponding correlation coefficient between past absolute daily returns and the weights placed on them by the CMM model. We also compute the coefficient's statistical significance. CMM overweights large (small) absolute returns, in line with an underreaction (overreaction) to news in the formation period, whenever the coefficient is positive (negative) and statistically significant at the 1%-level.

Fig. 3 shows the average fraction of the cross-section of stocks per year, for which we extract a weighting function consistent with a systematic underreaction to news in light blue and a systematic overreaction to news in black. For the remaining firm×month observations, the correlation between absolute past daily returns and weights is statistically indistinguishable from zero. The results are clear: throughout our out-of-sample period of 40 years, we find a clear dominance of CMM placing a larger weight on large absolute returns, in line with the idea that CMM exploits a persistent underreaction to news. For between 63% and 87% of firms does the model generate a positive correlation between past absolute daily returns and weights, consistent with this explanation. An overreaction, in contrast, is rare, being found for fewer than 10% of the stocks in the average month. For the remainder of the cross-section, we find that the correlation between absolute returns and weights is insignificant, such that the extracted CMM weighting function collapses to that of standard momentum.

To further ascertain that the dominant explanation for momentum-based profits in the market stems from an underreaction to large return days, we separately investigate the profitability of the CMM strategy for firms within each of the three buckets depicted in Fig. 3, capturing an under- (overreaction) to news, or an undecided verdict. The results are shown in Table 4. Consistent with the breakdown in Fig. 3 showing that CMM predominantly extracts a trend from an underreaction to news released in the formation period, its trading signal among these stocks is most valuable: the value-weighted high-minus-low quintile spread amounts to 1.19% per month, which is highly statistically significant. At the same time, we find that CMM has a harder time exploiting an overreaction. The high-minus-low return spread is not statistically

Table 4
CMM's performance and correlation.
The table shows time-series average portfolio returns. Stocks are first sorted by the correlation of their past absolute returns $|r|$ and weight w allocated by our model. Within each portfolio stocks are then again sorted into quintiles by CMM's signal. We continue to value-weight stocks with their market capitalization within each portfolio. *** (**, *) denotes that the return is statistically significant at the 1% (5%, 10%) level using HAC robust standard errors.

	CMM					
	LOW	2	3	4	HIGH	HL
Overreaction	0.59	0.81	0.31	0.55	0.49	−0.10
Undecided	0.52	0.38	0.67**	0.36	0.45	−0.06
Underreaction	−0.39	0.23	0.75**	0.94***	0.80***	1.19***

significant and slightly negative at −0.1% per month. Finally, we find corroborating evidence that assigning an equal importance to the returns realized in the formation period is sub-optimal. Whenever CMM's weighting of past returns collapses to that of standard momentum, evident in a correlation between absolute returns and weights that is indistinguishable from zero, the resulting high-minus-low return spread is slightly negative and insignificant at −0.06% per month. These results suggest that CMM learns to extrapolate from an underreaction to news released in the formation period, as its profitability concentrates in those stocks for which it places a greater weight on large absolute returns.

4.2. Risk-based explanations

Time-varying risk compensation. While our results up to this point favor a systematic underreaction to news as the driving force behind momentum-based profits in the stock market, CMM's signal may also capture time-variation in the compensation for risks. To this end, Kelly et al. (2021) show that the predictive power of momentum is subsumed by a measure of conditional risk exposure. Specifically, they estimate a conditional latent factor model of the form:

$$r_{i,t+1m} = \beta_{i,t}' f_{t+1m} + \varepsilon_{i,t+1m}, \qquad \mathbb{E}_t[r_{i,t+1m}] = \beta_{i,t}' \lambda_t, \tag{11}$$

and show that $\beta_{i,t}' \lambda_t$, i.e., the time-variation in the average compensation for systematic variation in the stock market, explains a sizable fraction of momentum returns.

We follow Kelly et al. (2021) and approximate the model in Eq. (11) using the instrumented principal component analysis (IPCA) put forth

Table 5
CMM and time-varying risk compensation.
The table shows panel regressions results. We report coefficient estimates and corresponding t-statistics using standard errors clustered by date. Coefficients for regressions using ranks are multiplied by 100 for readability.

	Raw				Ranks			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Const.	0.01	0.00	0.00	0.00	0.01	0.01	0.01	0.01
	(2.61)	(0.25)	(1.12)	(1.12)	(2.57)	(2.57)	(2.57)	(2.57)
CMM	0.09		0.06	0.06	3.72		2.27	2.23
	(14.33)		(9.68)	(9.59)	(15.63)		(8.00)	(7.85)
$\beta'\lambda$		0.80	0.50	0.50		3.69	2.18	2.13
		(16.25)	(11.70)	(12.16)		(16.67)	(8.50)	(7.41)
SM				−0.00				0.37
				(−0.13)				(0.88)

by Kelly et al. (2019) with five latent factors f[7]:

$$r_{i,t+1m} = \beta'_{i,t} f_{t+1m} + \epsilon_{i,t+1m}, \qquad \beta_{i,t} = z'_{i,t}\Gamma + \nu_{i,t}. \tag{12}$$

Here, βs are a linear function of observable firm characteristics z. We use the same set of 153 firm characteristics as in the estimation of the CMM model and estimate the IPCA model in a rolling out-of-sample fashion. Specifically, we use data up to month t to estimate the mapping between the firm characteristics and factor sensitivities, Γ_t, as well as factor realizations $f_{t+1,t}$:

$$\hat{f}_{t+1} = \left(\hat{\Gamma}'_t Z'_t Z_t \hat{\Gamma}_t\right)^{-1} \hat{\Gamma}'_t Z'_t r_{t+1}, \tag{13}$$

Here, Z_t denotes the matrix that stacks the characteristics of each firm i available at time t. The weights of each stock in the factor portfolios are known at time t, just like they are in standard portfolio sort-based factor construction. The estimated price of risk is the average of factor realizations in the estimation period, i.e., $\hat{\lambda}_t = \frac{1}{t}\sum_{\tau=0}^{t} \hat{f}_\tau$.

In panel regressions, we then investigate if the predictive power over future monthly excess stock returns of the CMM-based signal is subsumed by the time-varying compensation for risk, $\beta_{i,t}\lambda_t$:

$$r_{i,t+1m} = c_0 + c_1 \cdot \mathbb{E}_t^{\text{CMM}}[r_{i,t+1m}] + c_2 \cdot (\beta_{i,t}\lambda_t) + c_3 \cdot \mathbb{E}_t^{\text{SM}}[r_{i,t+1m}] + \epsilon_{i,t+1} \tag{14}$$

The results are provided in Table 5. As in Kelly et al. (2021), we separately show the results for the raw momentum signals in the columns to the left, and the cross-sectional percentage ranks of the signals in the columns to the right.

For both the raw signals and their cross-sectional ranks, we find in specification (1) that the CMM-based momentum signal predicts future stock returns well. Stocks with a higher CMM signal earn significantly larger returns over the next month. We next confirm in specification (2) the result of Kelly et al. (2021) that a description of the time-varying compensation for risks in the stock market also predicts future stock returns well: both the coefficient for the raw $\beta'\lambda$ signal and its cross-sectional ranks are large and statistically highly significant.[8] In specification (3), we combine both signals in a single regression. We find that while both coefficients decrease by about a third, compared to the univariate regressions, they retain their economic and statistical significance. Whereas the predictive power of standard momentum is well explained by it capturing time-variation in the compensation for risks in the stock market, the predictability of CMM's signal is only modestly impacted.[9] The absence of a clear relationship between time-varying risk compensation and CMM-based momentum profits in turn gives further credence to our empirical results in favor of a systematic underreaction to news giving rise to momentum in the stock market. Finally, in specification (4) we include the standard momentum signal as a control. Coefficients and their significance remain virtually unaffected, while the coefficient for standard momentum stands insignificant. This again underpins the previous finding that CMM subsumes standard momentum.

Momentum crashes. Daniel and Moskowitz (2016) propose an explanation for momentum profits in their inherent risk of so-called "momentum crashes" – brief investing periods that wipe out decades of previous profits. Our results in Table 1 and the visualization of the cumulative dollar value of investing in CMM in Fig. 4 suggest that CMM is not subject to the same severity of crashes as standard momentum is. In fact, the maximum drawdown for the full sample period is −27.6%, which compares favorably to the −80.3% lost by standard momentum. In turn, our results suggest that while crashes are a potential explanation for standard momentum, they do not serve as a comprehensive explanation for all types of momentum-based profits.

The results of the double sort in Table 2 furthermore show that the resilience of CMM to momentum crashes lies in the identification of stocks that are classified as "losers" by standard momentum, but that significantly outperform over the coming months. Among standard momentum losers, the CMM signal is most profitable, and generates a highly significant value-weighted high-minus-low decile spread of 1.7% a month.

4.3. Alternative explanations for momentum

In recent work, Goyal et al. (2024) show that the frog-in-the-pan hypothesis of Da et al. (2014) is the most likely explanation for momentum in international stock markets. Da et al. (2014) argue, and show empirically for the U.S., that investors underreact to discrete information releases. They measure the information discreteness (ID) as follows:

$$\text{ID}_{i,t} = \text{sign(PRET)} \times (\%\text{neg} - \%\text{pos}), \tag{15}$$

where %neg and %pos are the percentage of daily returns that are positive or negative, and PRET is the return over the past 11-months, that is, the standard momentum signal. Following the author's argument, a lower ID should therefore signal amplified momentum profits.

In addition, Goyal et al. (2024) consider various other empirical proxies to explain momentum profits. Following Daniel and Titman (1999), who hypothesize that the impact of investor overconfidence is strongest among firms whose intrinsic value is hard to determine, they use the book-to-market ratio of a firm as an observable proxy of overconfidence. Likewise, drawing on the work of Lee and Swaminathan (2000) and Odean (1998), they adopt turnover as a further indicator of overconfidence. The central hypothesis posits that a lower book-to-market ratio or higher turnover is associated with greater momentum profits.

Hong et al. (2000) hypothesize that the speed with which information is impounded into market prices is related to the number of financial analysts covering a firm. As larger firms are typically followed by more analysts, they cross-sectionally regress the log of one plus the number of analysts covering a firm on the its log market capitalization.

[7] We acknowledge the word of caution voiced in Kelly et al. (2019) that IPCA may not exclusively capture risk factors, but also systematic mispricing in the stock market. We follow Kelly et al. (2021) in estimating IPCA, as the authors highlight the improvements of IPCA over alternative ways to estimate conditional latent factor models, both in alleviating staleness in estimated betas and resolving the issue of misspecification in observable factors. As we find that $\beta'\lambda$, which potentially includes the influence of mispricing, cannot explain the predictability of CMM, we are reinforced in believing that time-variation in factor sensitivities cannot explain CMM's performance.

[8] Note that the size of the coefficients for raw signals are not comparable as CMM targets cross-sectionally standardized returns as explained in Section 2.

[9] We replicate the results of Kelly et al. (2021) that $\beta'\lambda$ subsumes the predictive power of the standard momentum signal in Table 12 of Appendix B.

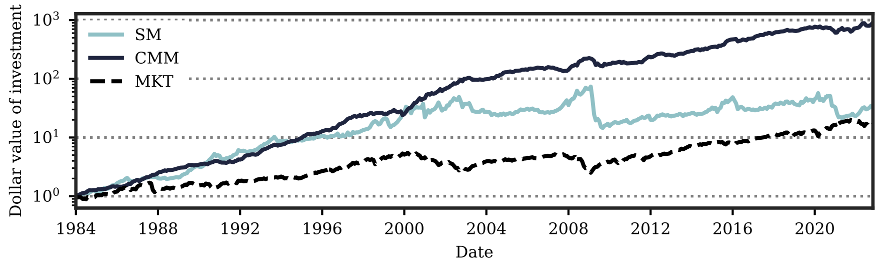

Fig. 4. Performance of Characteristic-Managed Momentum.
The figure shows the cumulative returns to the Characteristic-Managed Momentum strategy (CMM) and to standard momentum (SM) for the testing sample from 1983 through 2022 on a log scale. For comparison, we also add the cumulative returns of the market portfolio (MKT) to the graph.

The residual is then used as a proxy for the diffusion of information and a lower reading should signal higher momentum profits.

Tversky et al. (1982) show that subjects use anchors when making up their minds about unknown quantities. To this end, George and Hwang (2004) suggest that investors may use the 52-week high price of a stock as an anchor, as it is readily available. In this logic, investors perceive stocks with prices close to the 52-week high as expensive relative to stocks with prices further away from it, which in turn leads to a relative undervaluation of the former and a relative overvaluation of the latter. As a result, momentum profits should be negatively related to the proximity of the stock price to its 52-week high.

As a final proxy, Goyal et al. (2024) consider the stock's return volatility measured over the past 12 months, with a least 100 days of valid return data. An annualized return volatility greater than 300% is suspected to be a data error and set to missing. Sagi and Seasholes (2007) present a model in which the risk premium varies over time, with momentum as the compensation for a larger exposure to risk for past winners. Sagi and Seasholes (2007) note that in their model, momentum increases in the stock's return volatility.

As in Goyal et al. (2024), we investigate which of these proxies are related to the profitability of CMM through cross-sectional regressions:

$$r_{i,t+1m} = \gamma_{0,t} + \gamma_{1,t}\mathrm{CMM}_{i,t} + \gamma_{2,t}X_{i,t} + \gamma_{3,t}X_{i,t} \times \mathrm{CMM}_{i,t} + \epsilon_{i,t+1m}, \tag{16}$$

where X is one of the empirical proxies described above. We process the variables on the right-hand of the regression equation as in Goyal et al. (2024) and every month winsorize at the 0.5% and 99.5% level, and standardize across the entire sample to have zero mean and unit standard deviation. Furthermore, we exclude microcap stocks and drop the 3% smallest stocks by market capitalization, following Fama and French (2017). The predictors X are signed such that we would expect a positive interaction $X \times \mathrm{CMM}$ if the respective explanation of momentum for which X proxies is reflected in the return data. Returns are given in basis points, i.e., multiplied by 100.

The results are provided in Table 6. In Panel A, we show the results of cross-sectional regressions using one proxy X at a time. In Panel B, we run cross-sectional regressions with all predictors at once, to understand which survive the consideration of all other explanations. Consistent with the results presented so far, an increase in CMM significantly signals an increase in next month's returns. Across the different empirical proxies for explaining momentum profits, we find that the proximity to the 52-week high, the discreteness with which information entered the market (ID), and the stock's past return volatility all signal amplified CMM profits with the correct sign. In contrast, the impact of analyst coverage is insignificant, and the impact of both proxies for overconfidence – the book-to-market ratio and turnover – are significant but with the wrong sign, signaling lower CMM profits.

When we simultaneously consider all six proxies in Panel B of Table 6, we find that only the interaction between CMM's signal and ID remains statistically significant and produces the correct sign. The interactions with book-to-market ratio and turnover are both related to CMM's profits inversely to what the respective theories would predict. These results agree with the preferred explanation for standard momentum profits in international markets by Goyal et al. (2024) that investor underreaction to discrete information releases best agrees with the data. The results also agree with our previous results that CMM's profitability resides in the subsample of stocks for which CMM uncovers an underreaction to news provided (see Fig. 3 and Table 4), and that a risk-based explanation is an unlikely candidate (see Table 5).

5. Robustness

In this section, we start with showing the prevalence of characteristic-managed momentum in international equity markets, find convincing evidence of characteristic-managed *time-series momentum*, and analyze the impact of transaction costs on CMM's performance, showing that it remains profitable after adding half-spreads to each transaction. We also show that stock market factor models are unable to explain the returns produced by CMM, and that alternative augmentations of the standard momentum strategy, which have been proposed by the literature, fail to match CMM's performance.

International data. Hanauer and Windmüller (2023) show that the momentum effect is not limited to the U.S. stock market but also observable in emerging and other developed equity markets. We thus test if CMM also improves the momentum performance in other stock markets around the globe. For this, we estimate our model separately on data for emerging and developed markets. Since data coverage improves in the 1990s, we use all available data up to the beginning of 2003 to estimate the first model, such that first predictions start in January 2003. Once again, we refit the model every two years. Appendix C provides an overview of the 39 included countries and further outlines the filters applied and the model estimation approach. Table 7 shows CMM's performance next to standard momentum for the international data.

The results show that our approach generalizes to international markets. As before, we note that CMM not only improves substantially the strategies return but also significantly increases its Sharpe ratio. At the same time we find lower downside risk as measured by the maximum drawdown (MDD) across all specifications. Average returns stand at 19.7% a year with a Sharpe of 1.34 in non-U.S. stock markets, comfortably outperforming the standard international momentum strategy. Average returns are higher in developed markets at 26.7% per year (vs. 22.5% for emerging markets) but Sharpe ratios are higher in emerging markets (1.44 vs. 0.84 in developed markets). This evidence gives further validity to our approach of identifying modern and novel momentum signals that work across the globe.

Transaction costs. We use bid–ask spreads to estimate transaction costs for CMM. We subtract the prevailing half spread from the performance of the stocks that enter into the long and short leg of CMM, once when we enter into the position (at the end of month t) and once more when we unwind the position (at the end of $t+1$). To limit

Table 6

Fama–MacBeth regressions of future returns on CMM and explanatory variables.

The table shows the results of cross-sectionally regressing stock returns in the next month on CMM, various empirical proxies for explanations of momentum profits in X, and their interaction. We process the variables on the right-hand of the regression equation as in Goyal et al. (2024) and every month winsorize at the 0.5% and 99.5% level, and standardize across the entire sample to have zero mean and unit standard deviation. Furthermore, we exclude microcap stocks and drop the 3% smallest stocks by market capitalization, following Fama and French (2017). The predictors X are signed such that we would expect a positive interaction $X \times$ CMM if the respective explanation of momentum for which X proxies is reflected in the return data. Returns are given in basis points, i.e., multiplied by 100. Panel A considers one proxy at a time, and Panel B considers all proxy in one regression.

	–	−B/M	Turn	resAnly	−52wHi	−ID	RetVol
Panel A: Regressions with single predictor each.							
CMM	1.070	0.966	1.038	0.893	0.722	1.030	0.847
	(9.94)	(9.70)	(9.75)	(8.87)	(7.89)	(9.96)	(9.26)
X		−0.182	−0.228	−0.098	−0.190	−0.096	−0.039
		(−1.89)	(−2.79)	(−1.76)	(−1.28)	(−1.56)	(−0.23)
CMM × X		−0.153	−0.084	−0.026	0.291	0.175	0.148
		(−4.80)	(−2.42)	(−0.83)	(6.28)	(6.16)	(5.29)
Panel B: Regression with all predictors at once.							
CMM	0.819						
	(9.68)						
X		−0.039	−0.108	−0.107	−0.121	−0.018	0.054
		(−0.52)	(−2.09)	(−3.26)	(−1.18)	(−0.61)	(0.31)
CMM × X		−0.140	−0.101	0.007	0.021	0.089	0.042
		(−2.69)	(−3.23)	(0.24)	(0.59)	(3.95)	(0.91)

Table 7

International data.

The table compares the performance metrics for the Characteristic-Managed Momentum (CMM) and standard momentum (SM) strategy in the U.S. and across international markets. The sample is based on the data provided by Jensen et al. (2023) and includes only countries classified as either "developed" or "emerging". Appendix C presents a summary on the data and outlines our filter and model estimation approach. Results below span the out-of-sample time-period between February 2003 and December 2022. MDD denotes the maximum drawdown of the portfolio's value and CE($\gamma = 5$) denotes the certainty equivalent of a CRRA investor with a risk-aversion level of 5.

	US		Non-US		Developed		Emerging	
	CMM	SM	CMM	SM	CMM	SM	CMM	SM
$\bar{r}$	0.145	0.043	0.197	0.137	0.267	0.155	0.225	0.102
	(3.92)	(0.56)	(4.57)	(1.83)	(3.05)	(1.83)	(6.91)	(1.28)
$\sigma(r)$	0.140	0.289	0.147	0.285	0.317	0.354	0.157	0.298
SR	1.032	0.150	1.344	0.480	0.844	0.439	1.435	0.344
MDD	−0.369	−0.803	−0.174	−0.777	−0.243	−0.766	−0.234	−0.844
Skewness	−0.629	−1.607	3.844	−1.248	12.256	2.036	−0.346	−1.115
Kurtosis	2.223	6.542	41.947	5.922	175.494	29.077	0.724	3.762
CE($\gamma = 5$)	0.093	−0.307	0.153	−0.181	0.177	−0.287	0.162	−0.209

potential transaction costs ex ante, we focus our attention to those stocks with an observable bid–ask spread below or equal to 1% at the end of month t. This exercise has the benefit of assuring that CMM's profitability remains in this restricted universe of liquid stocks. To increase coverage, we use the spread estimates provided by Ardia et al. (2024) for the period prior to 1993. The results are shown in Table 8.

While the performance drops when considering transaction costs, CMM remains highly profitable both gross and net of costs. Gross of fees, CMM achieves an average return of 16.4% with a Sharpe ratio of about 1.2. Accounting for transaction costs reduces this return by 3.2% leading to a net return of 13.2% at a Sharpe ratio of 0.98.

Time-series momentum. Unlike traditional cross-sectional momentum strategies, Moskowitz et al. (2012) document a significant performance of a "time-series momentum" strategy, which involves going long in futures contracts with positive past returns and short in those with negative past returns. Although conceptually similar, time-series momentum differs from cross-sectional momentum in two key ways: (i) it typically takes positions in all assets simultaneously, and (ii) as noted by Goyal and Jegadeesh, 2018, it is generally not a zero-dollar investment, since the relative number of assets with positive and negative past returns can vary.

Table 8

Transaction costs.

The table shows performance metrics for the Characteristic-Managed Momentum (CMM) strategy if the investable universe is restricted to stocks with a month t bid–ask spread below or equal to 1%. We sort firms into deciles based on CMM's signal and weight stocks with their market capitalization when calculating portfolio returns. We account for the half spread once when we enter into the positions (at the end of month t) and once more when we unwind them (at the end of $t+1$). We show full sample metrics as well as results for the pre- and post-2003 period. To increase coverage, we use the spread estimates provided by Ardia et al. (2024) for the period prior to 1993.

	CMM: Bid–Ask-Spread ≤ 1%							
	$\bar{r}$	$\sigma(r)$	SR	MDD	Skewness	TCosts	$\bar{r}^{\text{Net}}$	SR^{Net}
All	0.164	0.134	1.221	−0.333	−0.354	0.032	0.132	0.979
Pre 2003	0.210	0.140	1.503	−0.333	−0.604	0.044	0.166	1.184
Post 2003	0.120	0.128	0.939	−0.267	−0.112	0.021	0.099	0.770

Table 9
CMM-augmented time-series stock momentum.
The table shows results for a CMM-based time-series momentum strategy with different weighting schemes. Following (Lim et al., 2018), we examine the profitability of a dollar-neutral time-series version of our CMM strategy in the U.S. equity market. Following the authors, we form two portfolios based on the sign of either (i) each stock's return over the $t-12$ to $t-2$ formation window (SM), or (ii) the sign of CMM's prediction (TS-CMM). We go long stocks with a positive sign and short stocks with a negative sign. We include three different weighting schemes: (a) value-weighting based on market capitalization, (b) equal weighting, and (c) inverse volatility weighting using realized daily volatility over the formation period. To address the practical challenge of implementing time-series momentum in a universe as large as the U.S. stock market, we also follow (Lim et al., 2018) in testing a *dual-momentum* strategy. This approach first groups stocks by the sign of the return forecast, then forms quintile portfolios within each group. The final strategy goes long the top quintile of the winner group and short the bottom quintile of the loser group. MDD denotes the maximum drawdown of the portfolio's value and CE($\gamma = 5$) denotes the certainty equivalent of a CRRA investor with a risk-aversion level of 5.

	Value-weighting				Equal-weighting		Volatility-weighting	
	CMM	SM	Dual-CMM	Dual-SM	CMM	SM	CMM	SM
$\bar{r}$	0.087	0.063	0.262	0.247	0.193	0.094	0.137	0.084
	(4.82)	(3.77)	(7.49)	(5.02)	(8.22)	(4.61)	(7.04)	(4.59)
$\sigma(r)$	0.090	0.111	0.171	0.312	0.091	0.123	0.082	0.102
SR	0.969	0.570	1.535	0.791	2.113	0.768	1.668	0.819
MDD	−0.267	−0.308	−0.313	−0.779	−0.299	−0.422	−0.279	−0.393
Skewness	−0.018	−0.295	0.237	−1.028	−1.218	−1.806	−0.806	−1.375
Kurtosis	5.673	2.610	3.387	5.110	10.711	12.676	7.906	9.131
CE($\gamma = 5$)	0.067	0.032	0.190	−0.147	0.171	0.049	0.120	0.055

Following Lim et al. (2018), we examine the profitability of a time-series version of our CMM strategy in the U.S. equity market. The authors propose a dollar-neutral strategy by forming two portfolios based on the sign of each stock's return over the $t-12$ to $t-2$ formation window. Stocks with positive past returns are included in the winner portfolio, while those with negative returns comprise the loser portfolio. The strategy then takes a long position in the winner portfolio and a short position in the loser portfolio. While this design ensures dollar neutrality, the number of stocks in each leg may differ. For CMM-enhanced time-series momentum (TS-CMM), we sort stocks with a positive CMM return forecast into the winner and stocks with a negative forecast into the loser portfolio.

To construct the portfolios, Lim et al. (2018) test three weighting schemes: (a) value-weighting based on market capitalization, (b) equal weighting, and (c) inverse volatility weighting using realized daily volatility over the formation period. To address the practical challenge of implementing time-series momentum in a universe as large as the U.S. stock market, the authors also propose a *dual-momentum* strategy, combining elements of time-series and cross-sectional momentum. This approach first groups stocks by the sign of their past return, then forms quintile portfolios within each group. The final strategy goes long the top quintile of the winner group and short the bottom quintile of the loser group. We benchmark our CMM-based time-series momentum strategy against the standard time-series momentum approach, with results reported in Table 9.

Focusing on the value-weighted results, we find that characteristic-managed time-series momentum averages 8.7% a year at a Sharpe of 1, which easily outperforms standard time-series stock momentum with a return of 6.3% and a Sharpe of 0.57. Incorporating both the CMM-augmented time-series and cross-sectional momentum signals following Lim et al. (2018) further boosts the overall performance to 26.2% a year at a Sharpe of 1.5. Equal- and inverse volatility-weighted results are comparable.

Fixing $\hat{z}$. The scalar $\hat{z}$ in Eq. (8) governs how much CMM overweights positive vs. negative returns.[10] The calculation of $\hat{z}$ is both firm-specific and potentially changes over time, as it is calculated from firm characteristics, which naturally evolve across both dimensions. To understand how important allowing for this heterogeneity is, we show in Fig. 5 average returns and Sharpe ratios over our sample period for simple momentum strategies with exponential weighting and a fixed $\hat{z}$. For small negative values of $\hat{z}$ we find slightly higher returns and Sharpe ratios than for standard momentum (for which $\hat{z} = 0$), suggesting that momentum can in-sample be marginally improved when slightly overweighting negative returns in the formation period. Compared to our preferred formulation of CMM with $\hat{z}$ that varies over firms and over time, however, fixed-$\hat{z}$ variants underperform considerably. CMM's performance is not merely the result of a fixed weighting scheme of formation period returns that differs from that of standard momentum, but requires cross-sectional heterogeneity in how large positive and negative formation period returns are aggregated.

Factor exposure. We find no evidence that CMM's exposure to common factors explains its performance. Appendix D shows that the strategy produces economically large and statistically significant alphas for the Fama and French (2015) five factor model augmented by standard momentum, the q-factor model by Hou et al. (2015), and the behavioral factor model of Daniel et al. (2020). alphas range between 1.1% for the q-factor model to 1.4% for the behavioral factor model.

Spanning regressions. To support the double sort presented in Table 2, we conduct additional spanning regressions with results outlined in Table 10. The regressions support the evidence that CMM fully subsumes standard momentum but not vice versa. While the regression coefficient of CMM on the standard momentum portfolio is significantly positive – indicating that CMM captures a form of momentum – the intercept also remains significantly positive. On the flip side, the results indicate that the standard momentum factor is fully subsumed by CMM in return. This suggests that CMM contains additional information beyond what is captured by the standard momentum factor, consistent with the view that CMM represents a superior momentum signal.

We also conduct spanning regressions on a long-short portfolio generated by a neural network in the style of Gu et al. (2020) (GKX). Similar to CMM, the network architecture consists of three hidden layers and is trained directly on all 153 characteristics without imposing any restrictions. The training and estimation procedures follow

[10] Note that this does not imply that the performance of CMM is mechanically driven to stem from an underreaction to news. To illustrate this, consider a scenario where news – and thus returns – are on average positive. In this case, assigning a negative $\hat{z}$ would cause the model to overweight small and negative returns. According to the framework described in Section 4, this pattern would actually be interpreted as an overreaction to news, not an underreaction.

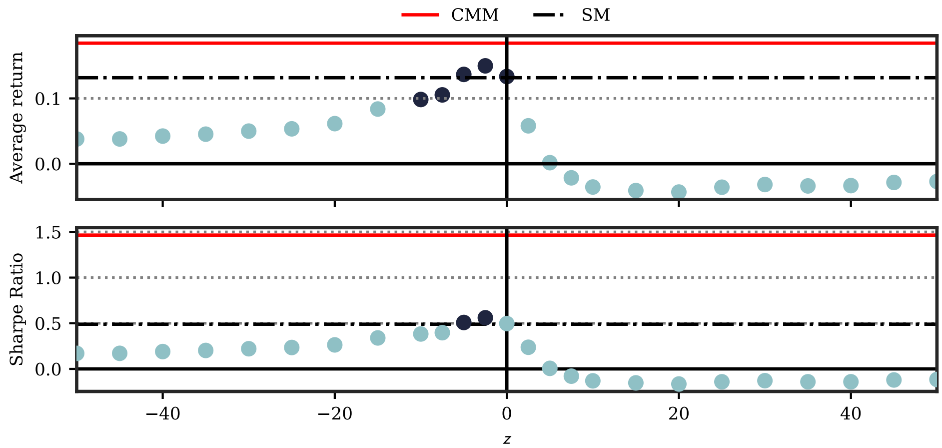

Fig. 5. Performance for Different but Fixed Values of $\hat{z}$.
The figure shows average returns (top) and Sharpe ratios (bottom) for simple momentum strategies with exponential weighting of formation period returns and a fixed scaling parameter $\hat{z}$. Dark blue circles denote that the average return (Sharpe ratio) is significant at the 5%-level using Newey–West standard errors (and the test by Lo, 2002).

Table 10
Spanning regressions.
The table shows the results of spanning regressions between CMM, standard momentum, and a long-short portfolio generated by a neural network in style of Gu et al. (2020). We use their best-performing predictive model, which is a neural network with 3 hidden layers (NN3), trained directly on the 153 characteristics which the authors find to perform best on a return prediction task. t-values are given in parentheses and based on Newey and West (1987) standard errors.

	CMM			MOM			GKX		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
const.	0.014	0.007	0.007	0.001	0.001	−0.000	0.016	0.006	0.006
	(7.53)	(4.97)	(5.07)	(0.36)	(0.21)	(−0.09)	(5.41)	(2.94)	(2.87)
MOM	0.191		0.085				0.236		0.109
	(2.99)		(1.41)				(2.14)		(1.16)
GKX		0.473	0.453	0.242		0.158			
		(14.28)	(11.96)	(2.65)		(1.32)			
CMM					0.288	0.178		0.692	0.660
					(2.41)	(1.21)		(9.23)	(10.51)
R^2	0.055	0.327	0.338	0.057	0.055	0.071	0.057	0.327	0.339

the same approach used for CMM. Consistent with Gu et al. (2020), who find that price trend variables, such as momentum and short-term reversal, are important predictors of future stock returns, we observe a positive covariance with the GKX-style machine learning portfolio. Nonetheless, the regression intercept remains significantly positive at 0.7% even in this specification, supporting the view that CMM represents an incremental signal.

Once we regress CMM's returns on both the standard momentum and GKX returns, we find that the coefficient for standard momentum drops from 0.191 to 0.085 and loses its statistical significance. This, however, does not mean that CMM is unrelated to momentum, but that much of the predictability of standard momentum is already captured by the GXK returns. Not only does the paper by Gu et al. (2020) find that the most important predictors of the associated strategy are based on different price trend and momentum signals, but Table 10 also shows that the GKX returns leave no alpha of momentum unexplained.

Model variations. We try to shed some light on the variable importance in Appendix E. For this we re-estimate the model while limiting its information set. We include a model which is limited to accounting (Acc. CMM), market (Market CMM) or hybrid (Hybrid CMM) characteristics. The latter contains characteristics that are constructed from both accounting and market information (such as the book-to-market ratio). We also include a model that is restricted to the information of the past daily returns (Ret. CMM). We find that a model relying solely on past daily returns fails to deliver a performance exceeding standard momentum. Incorporating firm characteristics is essential to inform how historical returns should be weighted. Once the model is provided with these characteristics, it outperforms the standard momentum benchmark. Notably, we show that both market and accounting information are required to uncover CMM's performance but crucially that accounting information on its own is more informative. This is contrary to the findings of Cakici and Zaremba (2023), who report a dominant role for market data in standard machine-learning predictions. Furthermore, market and accounting data appear to capture distinct aspects of firm behavior, as neither type alone matches the performance of the combined model. This suggests that it is not individual characteristics, but rather the full set of information and their interactions, that drive the model's success. This in turn strengthens our hypothesis that knowledge about the firm is important for uncovering differences in how past returns should be weighted.

Alternative augmentations of momentum. As a final robustness check, we compare CMM's performance improvement over standard momentum to that of other strategies based on different signal definitions proposed in the literature. We consider momentum strategies based on intermediate horizon past performance (MOM_12_7) and recent past performance (MOM_6_1), following Novy-Marx (2012) as well as the price-to-high momentum measure by George and Hwang (2004). The results are shown in Appendix F. None of these signals achieve a performance comparable to that of CMM.

Long-term performance. Our main results on the empirical drivers of CMM's success suggest that a systematic underreaction to news is a likely explanation for it. Fig. 6 in Appendix G shows the returns of the value-weighted high-minus-low decile portfolio based on CMM, when the expected return signal is generated at the end of month t and the portfolio return is evaluated in month $t + \tau$ with $\tau \in [1, 24]$ months. We find a gradual decline in the average realized returns, as we increase τ. A systematic overreaction to news released in the formation period should lead to an initially positive performance of CMM but a correction of this return pattern in the long-run, as market prices align with fundamentals.[11]

[11] In untabulated results, we confirm that the returns converge to and stay at zero for $\tau \in [25, 60]$ months.

6. Conclusion

While the traditional momentum strategy assumes that returns realized in the formation period are of equal importance, we show that allowing for a flexible aggregation of these returns generates significantly more profitable momentum signals. The resulting characteristic-managed momentum strategy subsumes the performance of standard momentum, continues to perform well in recent decades, and is resistant to momentum crashes, as it correctly identifies those previous losers that outperform in the future.

We find that the weights produced by our simple machine learning framework are sparse: just 35 of the 231 daily returns observed in the formation period already receive more than 50% of the total weight on average. Earnings announcements, days of market-wide jumps, and days with particularly large stock-specific returns are most informative.

Finally, we show that the correlation between the weights and absolute daily returns informs us about whether an over- or an underreaction to the information released throughout the formation period is a more likely candidate for explaining the success of momentum strategies in the stock market. Our results strongly favor an underreaction to news as the dominant driver of CMM's success: the strategy exploits an underreaction to news for between 60%–90% of stocks and only within this subsample do we find a statistically significant and economically meaningful value-weighted high-minus-low return spread. We furthermore find that CMM's profits are related to the discreteness with which information is released during the formation period (Da et al., 2014), for which Goyal et al. (2024) find reliable support for standard momentum profits in international stock markets. We perform several empirical tests to rule out alternative behavioral and risk-based explanations. Importantly, we show that CMM's profits are not explained by the variation in the compensation for risks in the stock market.

We ascertain the robustness of our results in international markets, by uncovering characteristic-managed time-series momentum, and in profitable returns net of transaction costs.

CRediT authorship contribution statement

Heiner Beckmeyer: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Timo Wiedemann:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

Authors declare that they have no relevant or material financial interests that relate to the research described in the paper.

Appendix A. Momentum signals from news and event days

Table 11
Momentum signals from news and event days
The table shows performance metrics for alternative momentum trading strategies based on different signals. Signals are based on the average return for different news days within the lookback period of $t-12$ and $t-1$ months. For earnings announcements (EA), GDP, CPI we consider a three day window around the news release date to account for announcement drifts and after-hour news releases. BAKER refers to large jumps in the stock market as defined by Baker et al. (2021) and $Q5$ considers the 5% days within the lookback-window with the highest absolute daily returns. Ensemble denotes an ensemble average of all signals except CMM. If there happen to be no news dates available for any stock, we set the signal to zero. In the table, MDD denotes the maximum drawdown of the portfolio's value and CE($\gamma=5$) denotes the certainty equivalent of a CRRA investor with a risk-aversion level of 5. All portfolios are value-weighted.

	CMM	EA	FOMC	GDP	CPI	BAKER	Q5	Ensemble
$\bar{r}$	0.185	0.041	0.060	0.033	0.077	−0.057	0.016	0.030
	(7.65)	(2.02)	(2.03)	(1.25)	(2.97)	(−1.91)	(0.48)	(0.84)
$\sigma(r)$	0.126	0.098	0.148	0.157	0.153	0.166	0.175	0.175
SR	1.465	0.421	0.406	0.209	0.502	−0.345	0.089	0.171
MDD	−0.276	−0.524	−0.567	−0.585	−0.493	−0.938	−0.826	−0.730
Skewness	−0.180	−0.132	0.569	−1.104	−1.164	−1.808	0.307	−0.702
Kurtosis	2.420	2.011	7.599	6.027	7.961	12.523	5.879	4.991
CE($\gamma=5$)	0.144	0.017	0.005	−0.039	0.008	−0.153	−0.063	−0.057

Appendix B. Momentum and time-varying risk compensation

Table 12
Momentum and time-varying risk compensation
The table shows univariate and bivariate panel regressions results (cf. (14)). We report coefficient estimates and corresponding t-statistics using standard errors clustered by date. Coefficients for regressions using ranks are multiplied by 100 for readability.

	Raw			Ranks		
	(1)	(2)	(3)	(1)	(2)	(3)
Const.	0.01	0.00	0.00	0.01	0.01	0.01
	(2.48)	(0.25)	(0.25)	(2.57)	(2.57)	(2.57)
SM	0.00		−0.00	1.30		0.51
	(0.61)		(−0.09)	(3.35)		(1.20)
$\beta'\lambda$		0.80	0.81		3.69	3.58
		(16.25)	(15.58)		(16.67)	(13.20)

Appendix C. International data

See Table 13.

Appendix D. Factor exposure

See Table 14.

Appendix E. Model variations

See Table 15.

Appendix F. Other augmentations of momentum signals

See Table 16.

Appendix G. Long-term performance

See Fig. 6.

Data availability

The authors do not have permission to share data.

Table 13
International Data
The table presents summary statistics for our international sample, based on data from Jensen et al. (2023). We restrict the sample to countries classified as either "developed" or "emerging", following the authors' classification criteria. Additionally, we apply their recommended screening filters: we include only one observation per security-month (**obs_main = 1**), restrict the sample to common stocks (**common = 1**), primary listings (**primary_sec = 1**), and stocks listed on prominent exchanges (**ex_main = 1**). We further limit the sample to country-month combinations where the country specific cross-section contains more than 100 firms. This leaves us with the 39 different countries shown below. Since data coverage improves in the 1990s, we use all available data up to the beginning of 2003 to estimate the first model. The first predictions begin in January 2003. As before, we re-estimate the model every two years, expanding the estimation window accordingly. To manage the large dataset, we estimate separate models for developed and emerging countries.

Country	MSCI Classification	Region	Total No. Firms	Min. No. Firms	Max. No. Firms	Start Predictions
ARE	Emerging	Middle east	129	100	110	2018-12-01
AUS	Developed	Asia	3246	761	1685	2003-01-01
BEL	Developed	Europe	267	121	153	2003-01-01
BRA	Emerging	South america	347	107	260	2008-04-01
CAN	Developed	North america	2603	769	1523	2003-01-01
CHE	Developed	Europe	412	210	249	2003-01-01
CHL	Emerging	South america	243	99	176	2003-05-01
CHN	Emerging	Asia	4664	1059	4443	2003-01-01
DEU	Developed	Europe	1663	658	961	2003-01-01
DNK	Developed	Europe	328	133	203	2003-01-01
EGY	Emerging	Middle east	257	101	207	2006-09-01
ESP	Developed	Europe	384	119	212	2003-01-01
FIN	Developed	Europe	250	115	163	2003-01-01
FRA	Developed	Europe	1448	629	756	2003-01-01
GBR	Developed	Europe	4229	1315	1952	2003-01-01
GRC	Emerging	Europe	388	138	316	2003-01-01
HKG	Developed	Asia	2795	614	2288	2003-01-01
IDN	Emerging	Asia	828	203	685	2003-01-01
IND	Emerging	Asia	5199	619	3857	2003-01-01
ISR	Developed	Middle east	788	170	494	2003-01-01
ITA	Developed	Europe	655	236	362	2003-01-01
JPN	Developed	Asia	5463	3433	3869	2003-01-01
KOR	Emerging	Asia	3274	1014	2341	2003-01-01
KWT	Emerging	Middle east	229	117	177	2007-01-01
MEX	Emerging	North america	145	100	123	2014-11-01
MYS	Emerging	Asia	1357	640	958	2003-01-01
NLD	Developed	Europe	270	100	178	2003-01-01
NOR	Developed	Europe	589	138	315	2003-01-01
NZL	Developed	Australasia	225	100	126	2004-08-01
PER	Emerging	South america	121	103	114	2018-09-01
PHL	Emerging	Asia	309	108	249	2003-01-01
POL	Emerging	Europe	1143	132	771	2003-01-01
SAU	Emerging	Middle east	228	106	208	2008-11-01
SGP	Developed	Asia	1067	400	682	2003-01-01
SWE	Developed	Europe	1290	259	788	2003-01-01
THA	Emerging	Asia	949	306	789	2003-01-01
TUR	Emerging	Europe	554	261	438	2003-01-01
TWN	Emerging	Asia	2520	651	2011	2003-01-01
ZAF	Emerging	Africa	607	226	319	2003-01-01

Table 14
Factor exposure
The table shows the results of regressing the returns of CMM on the five factors of Fama and French (2015) plus momentum in Panel A, the five factors of the q-factor model of Hou et al. (2015) in Panel B, and the three behavioral factors of Daniel et al. (2020) in Panel C. The factors are obtained from KennethFrench'swebsite, from KentDaniel'swebsite, and from global-q.org. t-values are given in parentheses and based on Newey and West (1987) standard errors.

	Panel A: FF5 + Momentum							
	α	MKT	SMB	HML	RMW	CMA	MOM	R^2
CMM	0.013	−0.005	−0.041	0.189	0.245	0.171	0.206	0.170
	(6.87)	(−0.17)	(−0.51)	(2.32)	(2.27)	(1.35)	(3.87)	
	Panel B: HXZ q-Factors							
	α	MKT^q	$Size^q$	IA^q	ROE^q	EG^q		R^2
CMM	0.011	0.006	0.100	0.365	0.289	0.304		0.162
	(5.16)	(0.15)	(1.13)	(3.45)	(2.82)	(2.41)		
	Panel C: DHS Behavioral Factors							
	α	MKT	PEAD	FIN				R^2
CMM	0.014	−0.024	0.048	0.262				0.100
	(6.82)	(−0.47)	(0.42)	(3.45)				

Table 15
Model variations
The table summarizes the performance metrics for the Characteristic-Managed Momentum (CMM) and standard momentum (SM) strategy. To assess which information is used in the modeling process to improve momentum signals, we add different model variations to the table. This includes a model which is limited to accounting (Acc. CMM), market (Market CMM) or hybrid (Hybrid CMM) characteristics. We also include a model that is restricted to the information of the past daily returns (Ret. CMM) only. MDD denotes the maximum drawdown of the portfolio's value and CE($\gamma = 5$) denotes the certainty equivalent of a CRRA investor with a risk-aversion level of 5.

	CMM	SM	Acc. CMM	Market CMM	Hybrid CMM	Ret. CMM
$\bar{r}$	0.185	0.132	0.183	0.137	0.117	0.063
	(7.65)	(2.81)	(6.09)	(4.25)	(3.15)	(2.49)
$\sigma(r)$	0.126	0.269	0.189	0.162	0.166	0.166
SR	1.465	0.490	0.965	0.844	0.704	0.383
MDD	−0.276	−0.803	−0.461	−0.626	−0.667	−0.481
Skewness	−0.180	−1.474	−0.131	−0.532	0.436	−0.626
Kurtosis	2.420	7.010	4.306	5.161	2.315	4.212
CE($\gamma = 5$)	0.144	−0.155	0.088	0.064	0.050	−0.013

Table 16
Other augmentations of momentum signals
The table shows full-sample performance metrics for momentum strategies based on different signals. We show results for the Characteristic-Managed Momentum (CMM) and for the standard momentum (SM) strategy. We also consider momentum strategies based on intermediate horizon past performance (MOM_12_7) and recent past performance (MOM_6_1), following Novy-Marx (2012), as well as the price-to-high momentum measure of George and Hwang (2004) (GH04). In terms of performance metrics, MDD denotes the maximum drawdown of the portfolio's value and CE($\gamma = 5$) denotes the certainty equivalent of a CRRA investor with a risk-aversion level of 5.

	CMM	SM	MOM_12_7	MOM_6_1	GH04
$\bar{r}$	0.185	0.132	0.105	0.092	0.023
	(7.65)	(2.81)	(2.71)	(2.35)	(0.81)
$\sigma(r)$	0.126	0.269	0.199	0.240	0.185
SR	1.465	0.490	0.527	0.382	0.122
MDD	−0.276	−0.803	−0.687	−0.812	−0.654
Skewness	−0.180	−1.474	−0.801	−1.415	−0.982
Kurtosis	2.420	7.010	3.647	9.202	6.195
CE($\gamma = 5$)	0.144	−0.155	−0.010	−0.137	−0.081

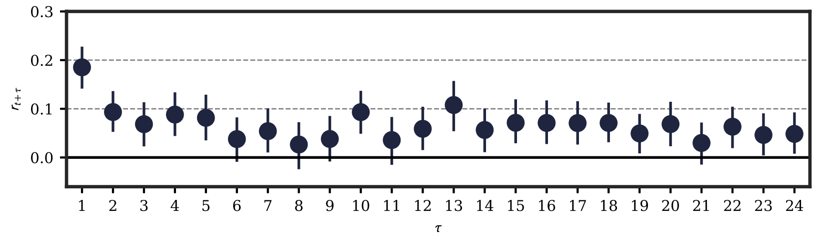

Fig. 6. Average returns of CMM formed on lagged signals.
The figure shows time-average returns of the CMM portfolio formed on τ-periods lagged signals. That is, we keep the signal generated in time t fixed and record returns of the CMM portfolio in month $t+\tau$, with $\tau \in [1, 24]$.

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